

Practice Quiz 01C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Using truth tables to test an argument.

1. An argument has premises $P \vee Q$ and $\neg P$ and conclusion Q . Make a complete four-row truth table and use it to determine whether the argument is valid. In your explanation, identify all rows in which both premises are true.

Solution

P	Q	$P \vee Q$	$\neg P$	Conclusion Q
F	F	F	T	F
F	T	T	T	T
T	F	T	F	F
T	T	T	F	T

The only row in which both premises are true is $P = F, Q = T$. The conclusion is true in that row, so there is no row with all premises true and conclusion false. Therefore the argument is valid. ■